7.2 Trigonometric Integrals

GitHub Copilot

January 22, 2024

1 Introduction

Trigonometric integrals are integrals that involve trigonometric functions. They often require techniques such as substitution, trigonometric identities, or integration by parts to solve.

2 Basic Trigonometric Integrals

The basic trigonometric integrals are:

$$\int \sin(x)\,dx = -\cos(x) + C$$

$$\int \cos(x)\,dx = \sin(x) + C$$

$$\int \tan(x)\,dx = -\ln|\cos(x)| + C$$

$$\int \cot(x)\,dx = \ln|\sin(x)| + C$$

$$\int \sec(x)\,dx = \ln|\sec(x) + \tan(x)| + C$$

$$\int \csc(x)\,dx = -\ln|\csc(x) + \cot(x)| + C$$

3 Strategies for Solving Trigonometric Integrals

When faced with a trigonometric integral, there are several strategies that can be used:

1. **Use a Pythagorean identity**: This can simplify the integral and make it easier to solve. 2. **Use a double-angle formula**: This can reduce the power of the trigonometric function. 3. **Use a half-angle formula**: This can also reduce the power of the trigonometric function. 4. **Use substitution**: This can simplify the integral and make it easier to solve. 5. **Use integration by parts**: This can be used when the integral is the product of two functions.

In the following sections, we will look at examples that demonstrate these strategies.

4 Example 1

Let's look at an example. Suppose we want to integrate $\int \cos^3(x)\,dx$.

For this integral, we can use the power-reducing identity $\cos^2(x) = 1 - \sin^2(x)$ to rewrite the integral as:

$$\begin{aligned}\int \cos^3(x)\,dx &= \int \cos(x)\cdot\cos^2(x)\,dx \\ &= \int \cos(x)\cdot(1-\sin^2(x))\,dx\end{aligned}$$

Now we can use substitution with $u = \sin(x)$ and $du = \cos(x)\,dx$:

$$\begin{aligned}\int \cos(x)\cdot(1-\sin^2(x))\,dx &= \int (1-u^2)\,du \\ &= u - \frac{1}{3}u^3 + C \\ &= \sin(x) - \frac{1}{3}\sin^3(x) + C\end{aligned}$$

So, $\int \cos^3(x)\,dx = \sin(x) - \frac{1}{3}\sin^3(x) + C$.

5 Example 2

Let's look at another example. Suppose we want to integrate $\int_0^\pi \sin^2(x)\,dx$.

For this integral, we can use the power-reducing identity $\sin^2(x) = \frac{1-\cos(2x)}{2}$ to rewrite the integral as:

$$\begin{aligned}\int_0^\pi \sin^2(x)\,dx &= \int_0^\pi \frac{1-\cos(2x)}{2}\,dx \\ &= \frac{1}{2}\int_0^\pi dx - \frac{1}{2}\int_0^\pi \cos(2x)\,dx\end{aligned}$$

The first integral is straightforward, and the second integral can be solved by using the basic integral of cosine:

$$\begin{aligned}\frac{1}{2}\int_0^\pi dx - \frac{1}{2}\int_0^\pi \cos(2x)\,dx &= \frac{1}{2}[x]_0^\pi - \frac{1}{2}[\frac{1}{2}\sin(2x)]_0^\pi \\ &= \frac{\pi}{2} - 0 \\ &= \frac{\pi}{2}\end{aligned}$$

So, $\int_0^\pi \sin^2(x)\,dx = \frac{\pi}{2}$.

6 Example 3

Let's look at another example. Suppose we want to integrate $\int \sin^4(x)\,dx$.

For this integral, we can use the half-angle formula $\sin^2(x) = \frac{1-\cos(2x)}{2}$ to rewrite the integral as:

$$\begin{aligned}\int \sin^4(x)\,dx &= \int \left(\frac{1-\cos(2x)}{2}\right)^2 dx \\ &= \frac{1}{4}\int (1 - 2\cos(2x) + \cos^2(2x))\,dx\end{aligned}$$

Now we can use the half-angle formula again to rewrite $\cos^2(2x)$ as $\frac{1+\cos(4x)}{2}$:

$$\begin{aligned}\frac{1}{4}\int (1 - 2\cos(2x) + \cos^2(2x))\,dx &= \frac{1}{4}\int (1 - 2\cos(2x) + \frac{1+\cos(4x)}{2})\,dx \\ &= \frac{1}{4}\int (1 - 2\cos(2x) + \frac{1}{2} + \frac{1}{2}\cos(4x))\,dx \\ &= \frac{1}{4}\int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2}\cos(4x)\right)\,dx\end{aligned}$$

The integral can now be solved by using the basic integrals of cosine:

$$\begin{aligned}\frac{1}{4}\int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2}\cos(4x)\right)\,dx &= \frac{1}{4}\left[\frac{3}{2}x - \sin(2x) + \frac{1}{8}\sin(4x)\right] + C \\ &= \frac{3}{8}x - \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C\end{aligned}$$

So, $\int \sin^4(x)\,dx = \frac{3}{8}x - \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C$.

Strategy for Evaluating $\int \sin^m x \cos^n x \, dx$

(a) If the power of cosine is odd ($n = 2k + 1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$$\int \sin^m x \cos^{2k+1} x \, dx = \int \sin^m x \, (\cos^2 x)^k \cos x \, dx$$
$$= \int \sin^m x \, (1 - \sin^2 x)^k \cos x \, dx$$

Then substitute $u = \sin x$. See Example 1.

(b) If the power of sine is odd ($m = 2k + 1$), save one sine factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine:

$$\int \sin^{2k+1} x \cos^n x \, dx = \int (\sin^2 x)^k \cos^n x \sin x \, dx$$
$$= \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx$$

Then substitute $u = \cos x$. See Example 2.

[Note that if the powers of both sine and cosine are odd, either (a) or (b) can be used.]

(c) If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \tfrac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \tfrac{1}{2}(1 + \cos 2x)$$

See Examples 3 and 4.

It is sometimes helpful to use the identity

$$\sin x \cos x = \tfrac{1}{2} \sin 2x$$

Figure 1:

Strategy for Evaluating $\int \tan^m x \sec^n x \, dx$

(a) If the power of secant is even ($n = 2k$, $k \geqslant 2$), save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$:

$$\int \tan^m x \sec^{2k} x \, dx = \int \tan^m x \, (\sec^2 x)^{k-1} \sec^2 x \, dx$$

$$= \int \tan^m x \, (1 + \tan^2 x)^{k-1} \sec^2 x \, dx$$

Then substitute $u = \tan x$. See Example 5.

(b) If the power of tangent is odd ($m = 2k + 1$), save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$:

$$\int \tan^{2k+1} x \sec^n x \, dx = \int (\tan^2 x)^k \sec^{n-1} x \sec x \tan x \, dx$$

$$= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x \, dx$$

Then substitute $u = \sec x$. See Example 6.

Figure 2:

2 To evaluate the integrals (a) $\int \sin mx \cos nx \, dx$, (b) $\int \sin mx \sin nx \, dx$, or (c) $\int \cos mx \cos nx \, dx$, use the corresponding identity:

(a) $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$

(b) $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$

(c) $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$

Figure 3: